Akademia Górniczo-Hutnicza
im. Stanisława Staszica w Krakowie

AGH University of Krakow

DIS 2026 report: HELP4CERN contributions and WG-summary visibility

Hamzeh Khanpour

AGH University of Krakow

(on behalf of HELP4CERN Study Group)

HELP4CERN collaboration report - DIS 2026, Bologna

Outline

1. DIS 2026 context
2. Four DIS 2026 HELP4CERN contributions
3. Visibility in DIS 2026 WG summaries
4. Feedback and next steps

DIS 2026 as a visibility point for HELP4CERN

Main message from DIS 2026

- The HELP4CERN programme was visible across several DIS 2026 working groups: machine concept, detector/staging, small-x QCD, and HET physics.
- The proposed LHμC was discussed as a possible high-energy lepton–hadron extension at CERN, not simply as an energy upgrade of the LHeC.
- The WG-summary slides gave useful external feedback on where the community sees the strongest physics and strategy questions.

Four DIS contributions reviewed

- K. Piotrzkowski: LHμC concept and programme.
- L. Forthomme: LHeC/FCC-eh detector and staging.
- P. Kotko: high-energy QCD and small-x reach.
- H. Khanpour: HET physics at a muon–hadron collider.

WG-summary visibility

- WG2: small-x and Q^2–x plane.
- WG3: HET physics prospects at LHμC.
- WG6: machine, detector and strategy outlook.

Report structure: from talks to next steps

What this report summarizes

- Four HELP4CERN-related DIS 2026 presentations.
- How these studies appeared in WG-summary material.
- Physics and technical feedback to sharpen for the white paper.
- ICHEP 2026 as the next opportunity for a coherent presentation.

DIS 2026 contribution pages: 289190, 289187, 289240, 289107.

DIS 2026 contribution: Krzysztof Piotrzkowski

Talk

LHμC: A high energy, high luminosity muon-hadron collider at CERN

K. Piotrzkowski, WG6 Current Upgrades and Future Experiments, 5 May 2026.

Physics and concept message

- Presented the LHμC concept based on reusing elements of the LHeC ERL infrastructure.
- Baseline example: a 500 GeV μ^+ beam colliding with the 7 TeV LHC proton beam, giving $\sqrt{s} \simeq 3.74$ TeV.
- Discussed the path from ultra-cold μ^+ production to booster/ring concepts and first machine–detector-interface issues.
- Framed the scientific programme from QCD and DIS to HET and representative BSM studies.

HELP4CERN relevance

- Provides the common accelerator and staging narrative for the white paper.
- Makes the physics–machine connection explicit: luminosity, energy, detector protection and feasibility.
- Connects LHμC to LHeC, muon-collider and FCC discussions.

DIS 2026 Indico contribution 289190

DIS 2026 contribution: Laurent Forthomme

Talk

A detector for top energy DIS

L. Forthomme, WG6 Current Upgrades and Future Experiments, 5 May 2026.

Detector and staging message

- Reviewed the status of detector concepts for top-energy DIS at the LHeC and extension toward FCC-eh.
- Used the LHeC idea of electron–proton/nucleus collisions at CERN with an ERL electron beam as the near-term reference.
- Discussed staged LHeC operation as a bridge from HL-LHC toward a possible later flagship collider.
- Highlighted synergies with ALICE 3, ePIC and future e^+e^- detector concepts.

HELP4CERN relevance

- Gives the detector realism needed for a credible lepton–hadron programme.
- Provides a reference geometry for later LHμC detector and MDI discussions.
- Helps separate near-term LHeC questions from more speculative LHμC extensions.

DIS 2026 Indico contribution 289187

DIS 2026 contribution: Piotr Kotko

Talk

Exploring the frontier of high energy QCD with the LHμC

P. Kotko, WG2 Small-x, Diffraction and Vector Mesons, 6 May 2026.

QCD message

- Presented LHμC as a high-energy DIS facility with access to extremely small Bjorken-x.
- Emphasized potential reach down to $x \sim 10^{-7}$ at perturbative virtualities around $Q^2 \simeq 1\ \mathrm{GeV}^2$.
- Motivated studies of gluon saturation, nonlinear QCD evolution and nuclear effects.
- Highlighted the complementary large-Q^2, low-x lever arm for PDF uncertainties and possible α_s extractions.

HELP4CERN relevance

- Supplies one of the clearest visual arguments for the LHμC: the extended Q^2–x plane.
- Connects the proposal naturally to EIC, HERA, LHeC and FCC-eh comparisons.
- Defines concrete QCD benchmarks for the white paper.

DIS 2026 Indico contribution 289240

DIS 2026 contribution: Hamzeh Khanpour

Talk

Perspectives for studies of the HET physics at the muon-hadron collider at CERN

H. Khanpour, WG3 Electroweak Physics, Top, Higgs and BSM, 7 May 2026.

HET message

- Presented benchmark inclusive cross sections for Higgs and key top-quark processes at LHμC working points.
- Highlighted opportunities for charged-current Higgs production and precision Higgs-coupling studies.
- Discussed a broader electroweak/top programme, including high-Q^2 sensitivity and rare-top directions.
- Included photon-induced processes and long-term muon–hadron options at FCC-hh energies.

HELP4CERN relevance

- Establishes the HET pillar of the LHμC physics case.
- Provides comparison points for HL-LHC, LHeC, FCC-ee and FCC-hh discussions.
- Identifies where dedicated simulations and coupling-fit assumptions must be sharpened.

DIS 2026 Indico contribution 289107

WG3 summary: LHμC HET physics highlighted

34

35

36

37

38

Muon-hadron collider at CERN: HET physics

H. Kahnpour - Perspectives for studies of the HET physics at the muon-hadron collider at CERN

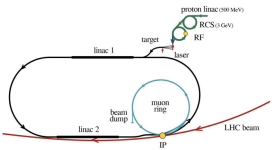
Combines the clean lepton-hadron initial state of DIS with a multi-TeV centre-of-mass energy, opening a new regime for Higgs, electroweak, top, and BSM studies

QCD & DIS frontier
Multi-TeV μp scattering extends the accessible (x, Q^2) plane well beyond HERA and the EIC/LHeC, strengthening the PDF, small-x, and α_s programme.

Top and rare-top physics
The top sector gives the clearest rate gain: already the 500 GeV LHμC benchmark enhances $t\bar{t}$ production by ~ 32 relative to the 50 GeV LHeC case, motivating precision top and FCNC studies.

Higgs–EW precision
Large CC/NC Higgs samples enable concrete coupling studies; the $h \to b\bar{b}$ benchmark reaches sub-percent sensitivity to g_{hbb}, while EW/SMEFT observables gain a larger high-Q^2 lever arm.

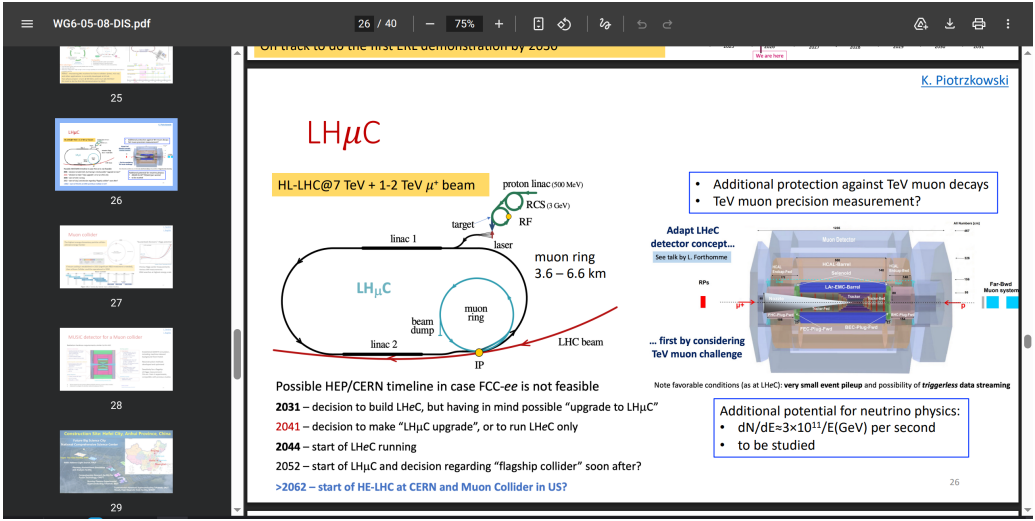
$\gamma\gamma$ and BSM portals
Photon-induced channels provide high-statistics $\tau^+\tau^-$ samples and high-mass WW, ZZ, $t\bar{t}$ probes, while direct μq and μg portals target RPV squarks, color-octet muons, and other muon-coupled BSM states.



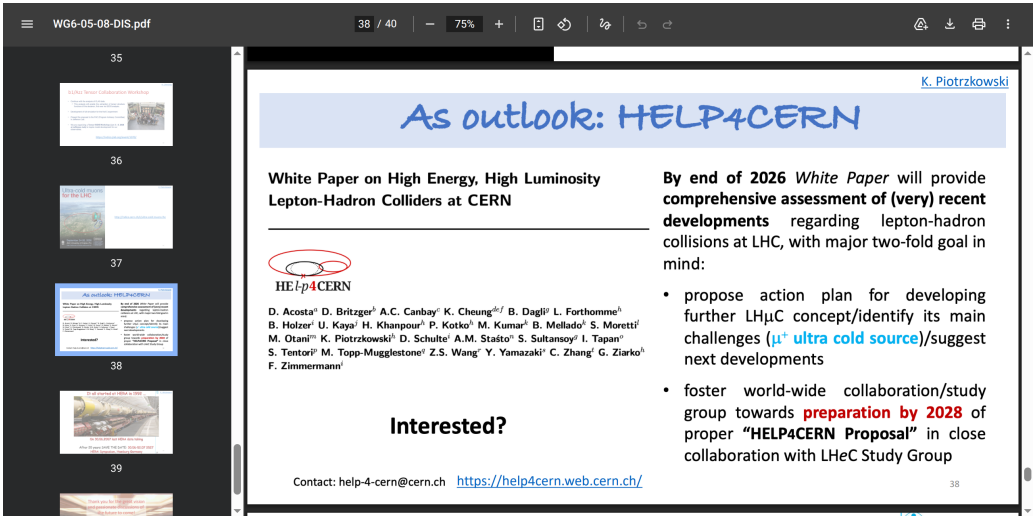
Conceptual layout of the proposed LHμC configuration.

37

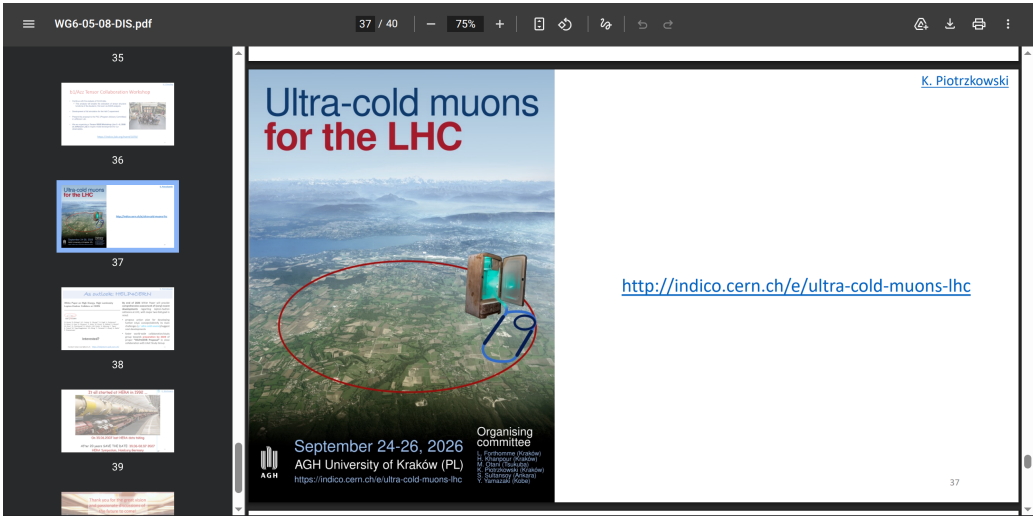
WG6 summary: LHμC concept and timeline



WG6 outlook: HELP4CERN white paper



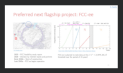
WG6 outlook: ultra-cold muons for the LHC



LHeC and LHeD detector concept

WG6-05-08-DIS.pdf 23 / 40 75%

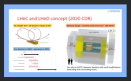
20



21



22



23



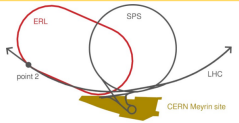
24



L. Forthomme

LHeC and LHeD concept (2020 CDR)

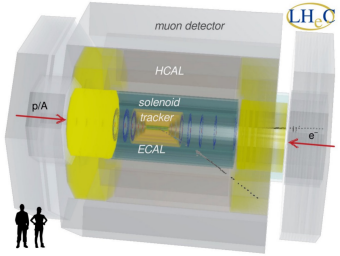
HL-LHC@7 TeV + 20-50 GeV e^- beam at IP2



E.g. decision in 2034 – operation in 2041



Modular design: "could be built tomorrow", 360 MCHF



Can rely on ALICE 3 detector baseline with small modifications (extending ECAL and adding HCAL)

23

Staged LHeC and later LHμC extension

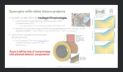
lhec_detector_forthomme_dis.pdf 13 / 18 75%


10


11

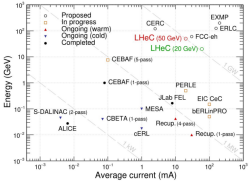

12


13


14

Staged LHeC

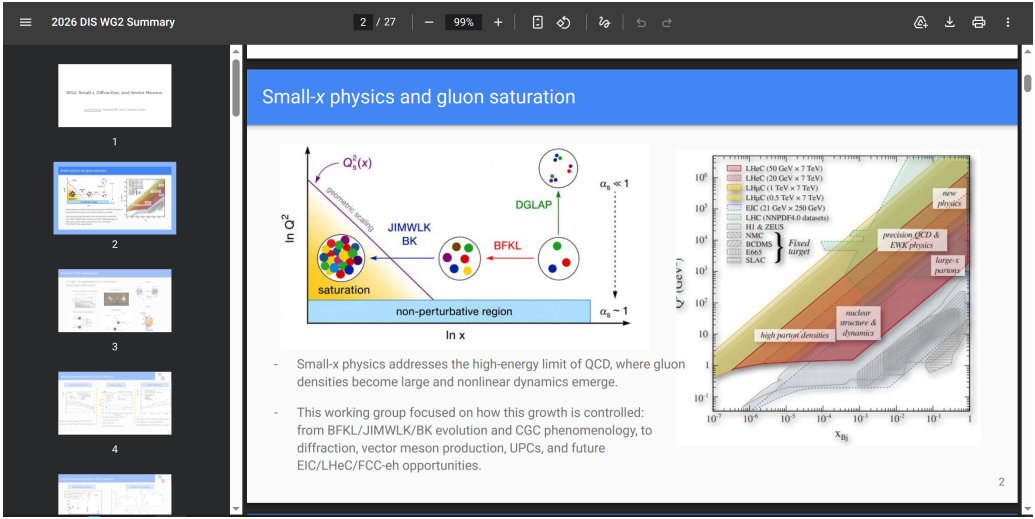
- Room for operating LHeC in steps, see e.g. low-energy "stage 1 LHeC" @ $\sqrt{s}$ = 0.75 TeV [arXiv:2503.20475 [hep-ex]/New J. Phys. 28 (2026) 013001]
  - single-pass ERL, dropping the need for splitting/recombination optics
  - increase of beam current (20 mA → 60 mA), still following the operational power constraints at highest-luminosity running mode (6×10^{33} cm^{-2}s^{-1})
- More than an experimental phase for e-h collisions at the LHC
  - can rely on ALICE 3 detector baseline with small modifications (HCAL & extension of ECAL for DIS, near-beam detectors instrumentation...)
  - physics case remains very competitive, altogether with HL-LHC projections: EW & top physics, BSM
  - ready for comparative eA/AA measurements at unprecedented scales



- After conventional LHeC operations, room for later extensions to **higher energy** lepton-hadron collisions, e.g. LHμC
  - See HELP4CERN talks: LHμC (K. Piotrzkowski, Tuesday@11:20), QCD@LHμC (P. Kotko, Wednesday@12:00), Higgs/EW/top@LHμC (H. Khanpour, Thursday@11:00)

laurent.forthomme@cern.ch DIS2026 — A detector for top-energy DIS 13

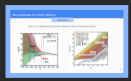
WG2 summary: small-x physics and saturation

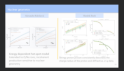


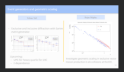
WG2 summary: new proposals for small-x physics


10


11


12

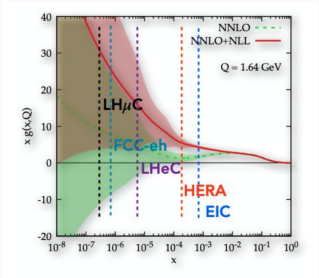

13

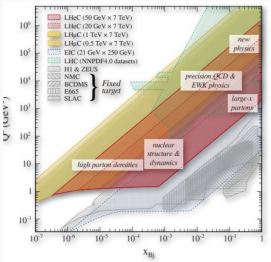

14

New proposals for small-x physics

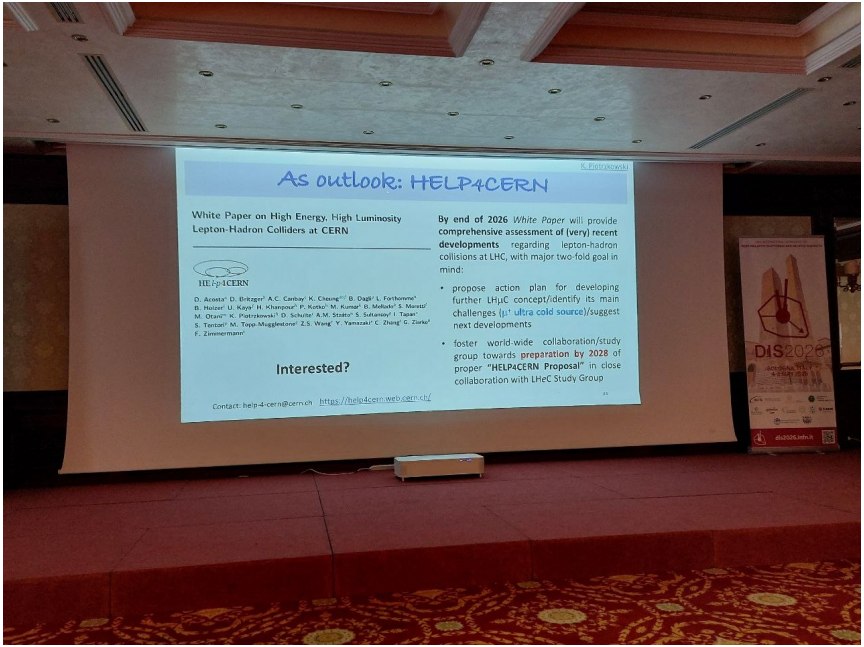
Piotr Kotko

LHμC is an opportunity for small-x physics in the not-so-distant future.





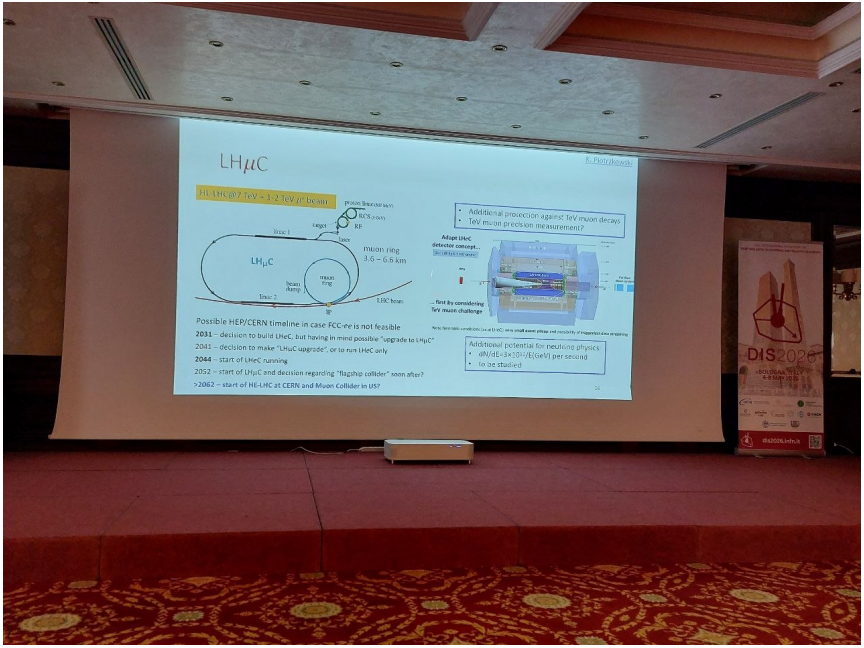
12



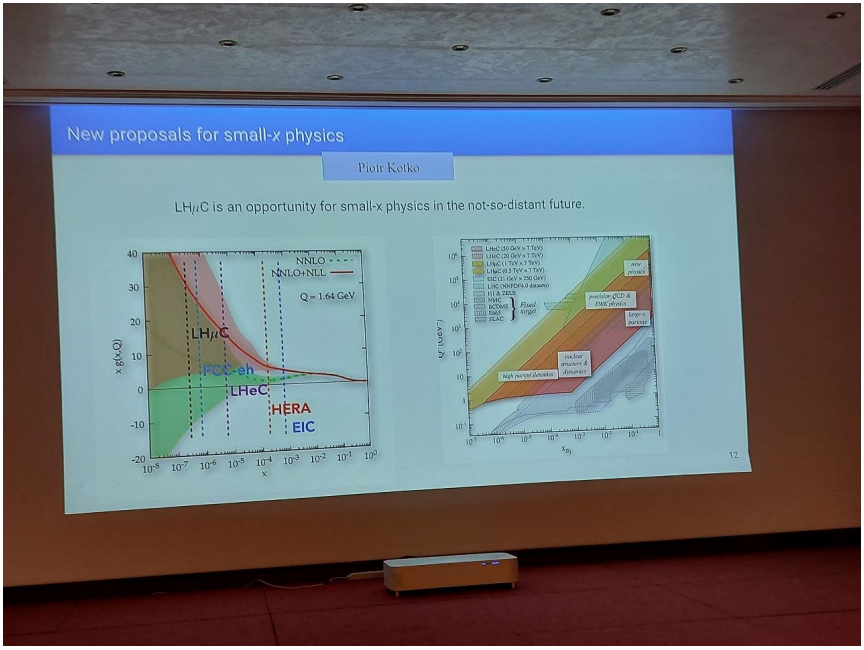
AGH
www.agh.edu.pl



AGH
www.agh.edu.pl



AGH
www.agh.edu.pl



AGH
www.agh.edu.pl

DIS 2026 feedback: physics interest and next questions

What attracted attention?

- The HELP4CERN proposal itself.
- The physics potential of future lepton–hadron collisions at CERN.
- The extended Q^2–x plane and small-x reach.
- The complementarity with EIC, HERA and LHeC.

Physics questions to sharpen

- Compare Higgs-coupling reach for HL-LHC, LHeC, LHμC and FCC-ee.
- Emphasize complementarity: CC Higgs, high-Q^2 DIS, top/EW, PDFs, photon-induced channels.

From DIS 2026 to ICHEP 2026

Status after DIS 2026

- The DIS talks were well received and represented in the WG summaries.
- All four ICHEP 2026 abstracts are approved.
- ICHEP is the next opportunity to present a coherent HELP4CERN/LHeC/LHμC programme.

Suggested ICHEP coordination

- Use one common introductory slide across talks: concept, staging and physics pillars.
- Align benchmark assumptions for $\sqrt{s}$, luminosity and comparison machines.
- Prepare one shared slide on technical challenges.

ICHEP 2026 contributions: 6986580, 6982821, 6983210, 6980628.


THANK
YOU!